

```

> with(DETools) :
> ecdif := diff(x(t), t) = -k·x(t) :
> cond_in := x(0) = x_0 :
> soll := dsolve({ecdif, cond_in}, x(t));

```

$$soll := x(t) = x_0 e^{-kt} \quad (1)$$

```

> y1 := unapply(rhs(soll), x_0, k, t);

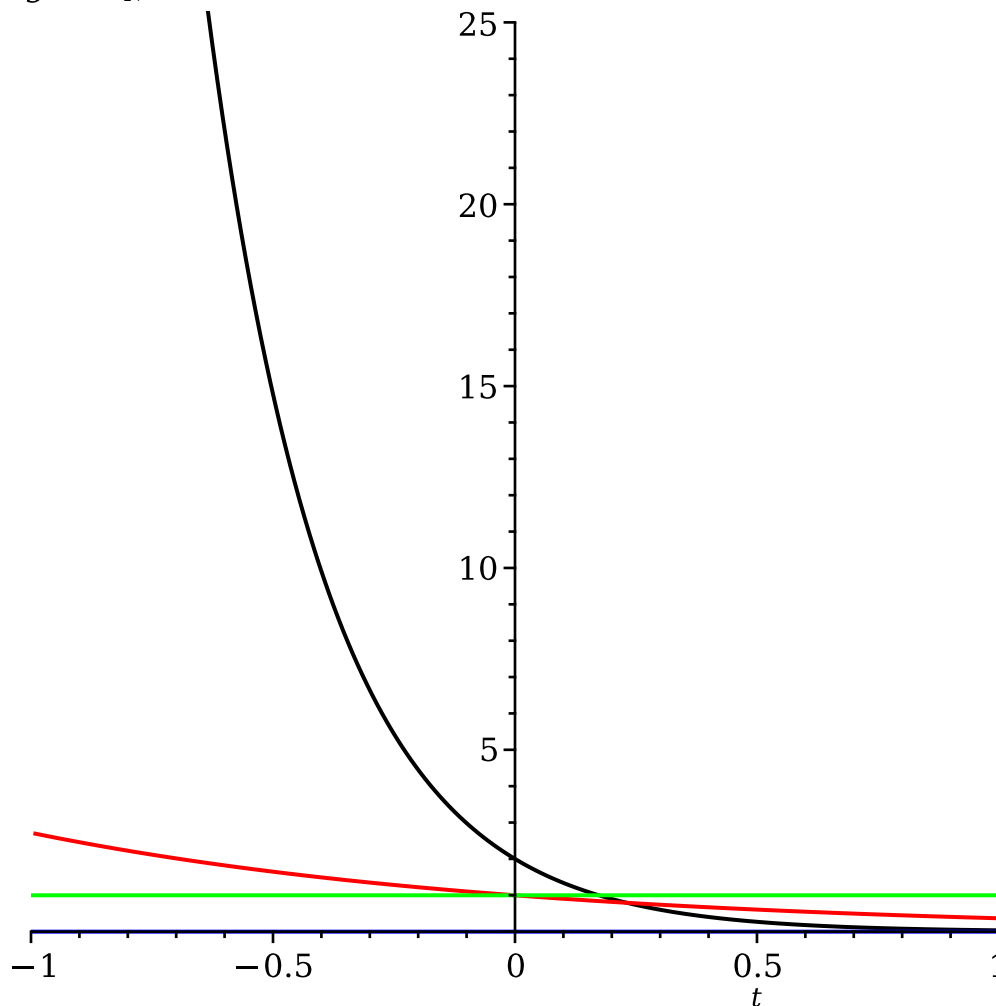
```

$$y1 := (x_0, k, t) \mapsto x_0 \cdot e^{-k \cdot t} \quad (2)$$

```

> plot([y1(2, 4, t), y1(1, 1, t), y1(0, 1, t), y1(1, 0, t)], t = -1..1, color = [black, red,
blue, green]);

```



```

> solve( (x_0/2 = y1(x_0, k, t), t);

```

$$\frac{\ln(2)}{k} \quad (3)$$

```

> evalf( solve( (x_0/2 = y1(x_0, k, 5730), k) );

```

$$0.0001209680943 \quad (4)$$

```

> k1 := solve(0.9 = y1(3, k, 2), k);

```

$$(5)$$

$$k1 := 0.6019864022 \quad (5)$$

$$> T_{12} := \text{evalf}\left(\frac{\ln(2)}{k1}\right);$$

$$T_{12} := 1.151433285 \quad (6)$$

>

> restart:

> with(DETools):

> ecdiff := diff(T(t), t) = -k·(T(t) - T_m):

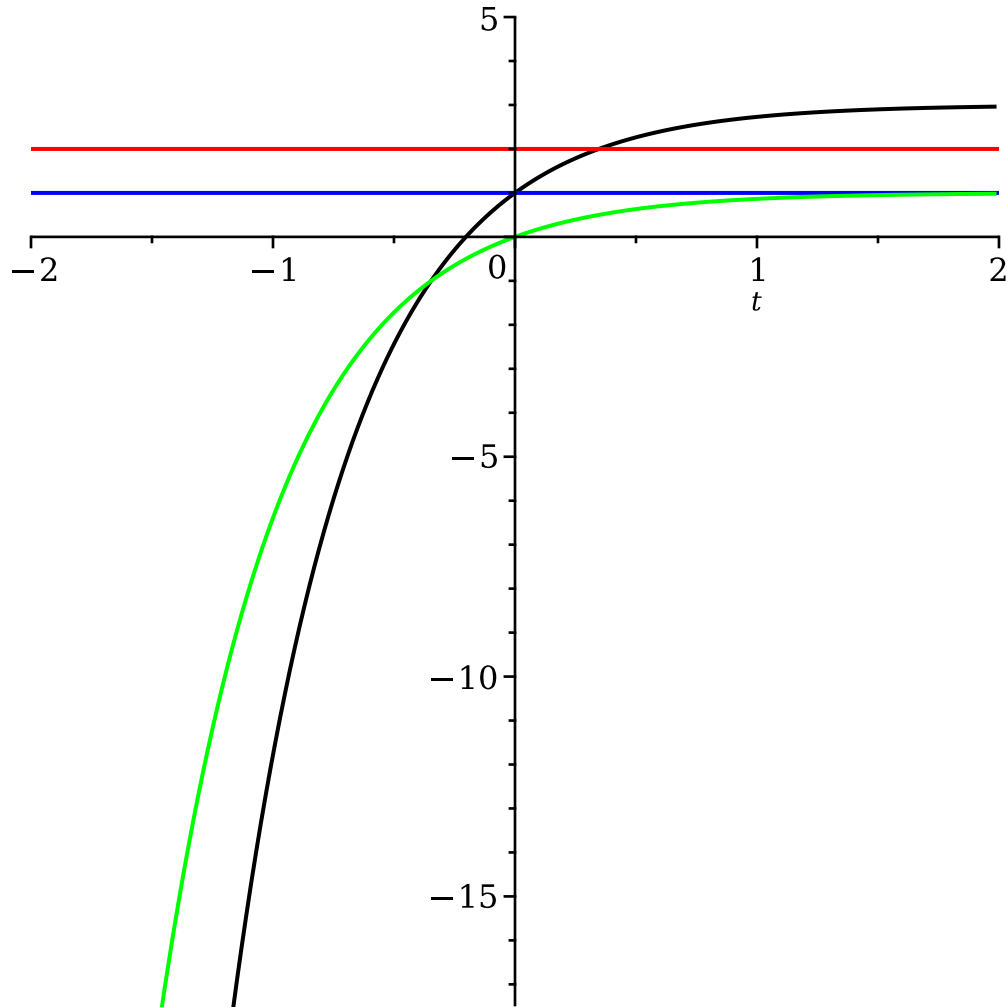
> cond_in := T(0) = T₀:

> soll := dsolve({ecdiff, cond_in}, T(t));

$$soll := T(t) = T_m + e^{-kt} (T_0 - T_m) \quad (7)$$

> y1 := unapply(rhs(soll), t, k, T₀, T_m):

> plot([y1(t, 2, 1, 3), y1(t, 3, 2, 2), y1(t, 5, 1, 1), y1(t, 2, 0, 1)], t = -2..2, color = [black, red, blue, green]);



> solve({y1(t, k, 36, 21) = 34.22, y1(t + 1, k, 36, 21) = 34.11}, {t, k});

$$\{k = 0.008355536648, t = 15.11804352\} \quad (8)$$

> restart:

```

> with(DETools) :
> ecdif1 := diff(x(t), t) = r·x(t) :
> cond_in := x(0) = x0 :
> ecdif2 := diff(x(t), t) = r0·x(t)· $\left(1 - \frac{x(t)}{K}\right)$  :
> soll1 := dsolve({ecdif1, cond_in}, x(t));
       $soll1 := x(t) = x_0 e^{rt}$ 

```

(9)

```

> soll2 := dsolve({ecdif2, cond_in}, x(t));
       $soll2 := x(t) = \frac{K x_0}{(K - x_0) e^{-r_0 t} + x_0}$ 

```

(10)

```

> y1 := unapply(rhs(soll1), x0, r, t);
       $y1 := (x_0, r, t) \mapsto x_0 \cdot e^{rt}$ 

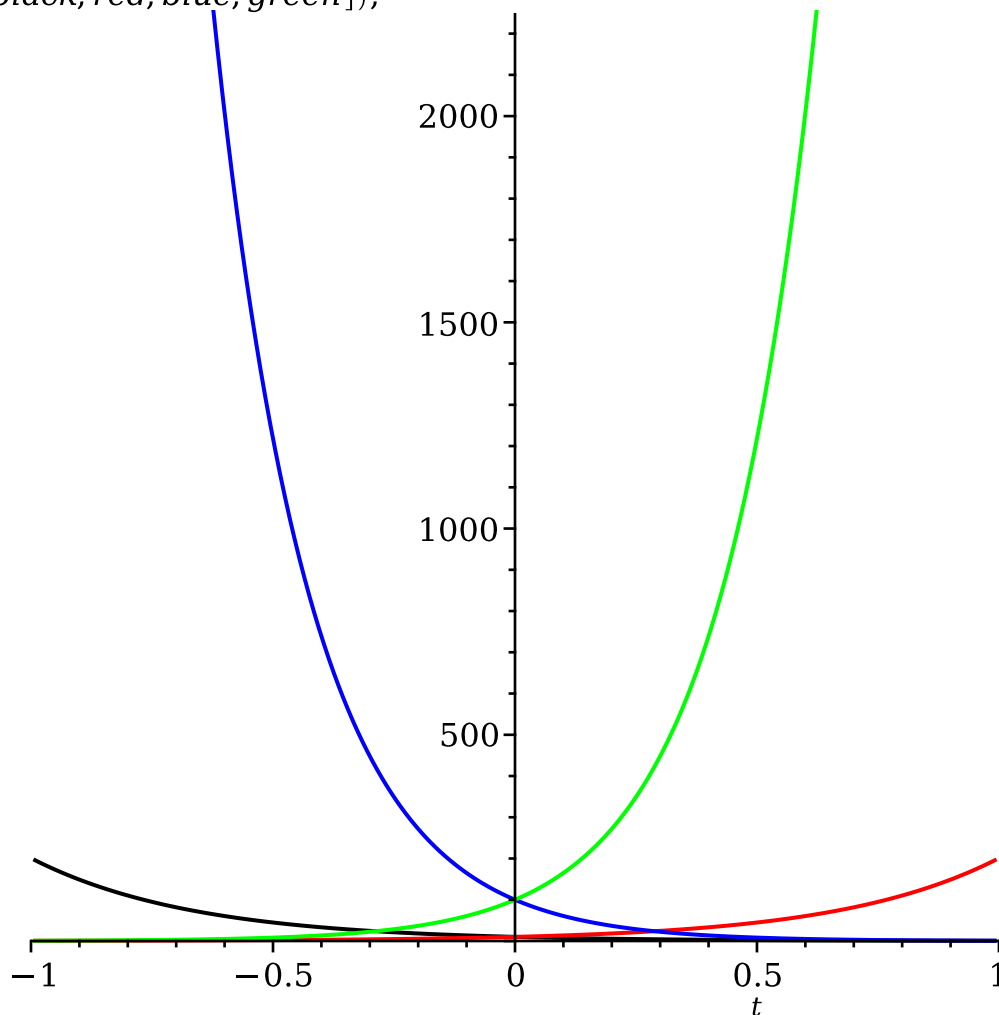
```

(11)

```

> plot([y1(10, -3, t), y1(10, 3, t), y1(100, -5, t), y1(100, 5, t)], t = -1 .. 1, color
      = [black, red, blue, green]);

```



```

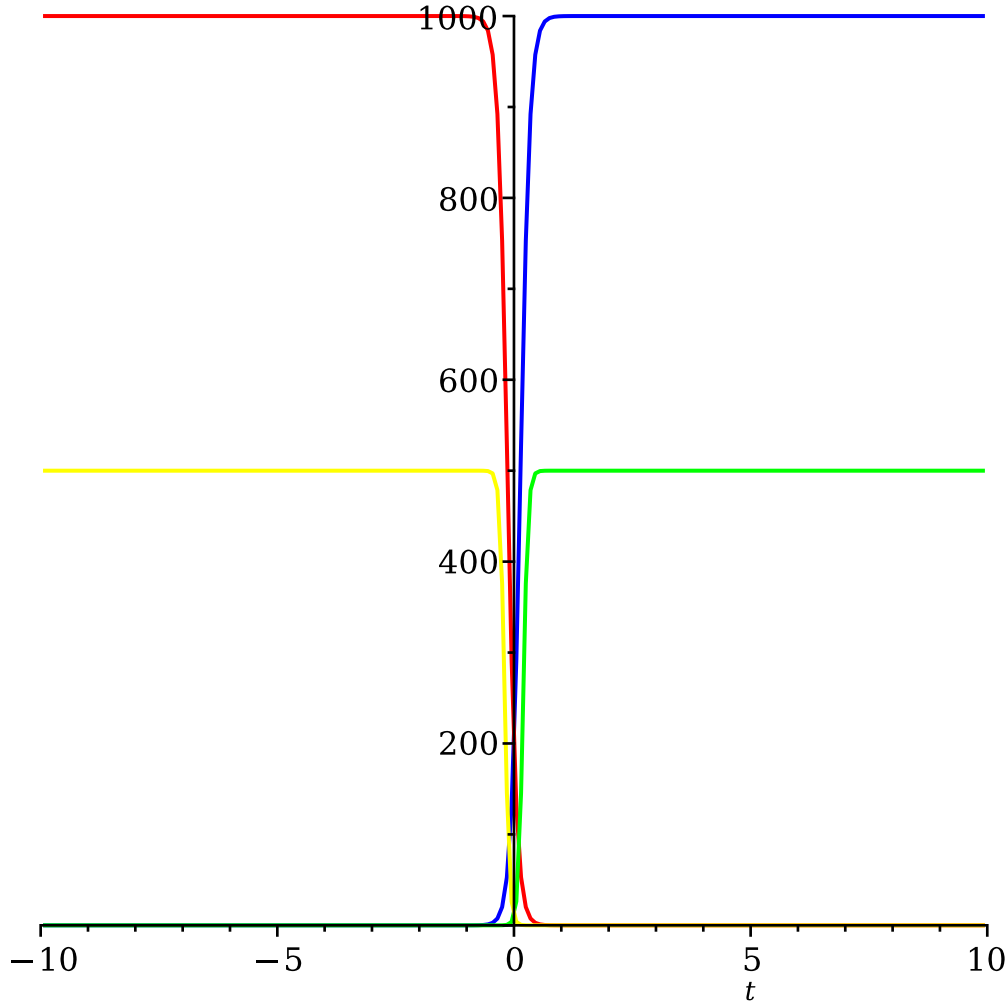
> y2 := unapply(rhs(soll2), K, x0, r0, t);

```

(12)

$$y2 := (K, x_0, r_0, t) \mapsto \frac{K \cdot x_0}{(K - x_0) \cdot e^{\frac{-r_0 \cdot t}{x_0}} + x_0} \quad (12)$$

```
> plot([y2(1000, 200, 10, t), y2(1000, 200, -10, t), y2(500, 10, 20, t), y2(500, 10, -20, t)], t = -10 .. 10, color = [blue, red, green, yellow]);
```



```
> rt := solve( {y1(9.6·106, r, t) = 9.6·106, y1(9.6·106, r, t + 10) = 12.9·106 }, {r, t});
```

$$rt := \{r = 0.02954642129, t = 0.\} \quad (13)$$

```
> r1 := rhs(rt[1]);
```

$$r1 := 0.02954642129 \quad (14)$$

```
> evalf(y1(9.6·106, r1, 110))
```

$$2.476135935 \times 10^8 \quad (15)$$

```
> rt1 := solve( {y2(K, 9.6·106, r0, t) = 9.6·106, y2(K, 9.6·106, r0, t + 10) = 12.9·106, y2(K, 9.6·106, r0, t + 20) = 17.1·106 }, {K, r0, t});
```

$$rt1 := \{K = 9.2364000 \times 10^7, r_0 = 0.03361533082, t = 0.\} \quad (16)$$

```
> K1 := rhs(rt1[1]);
```

$$K1 := 9.2364000 \times 10^7 \quad (17)$$

```
> r2 := rhs(rt1[2]);
```

	$r2 := 0.03361533082$	(18)
	$\rightarrow \text{evalf}(y2(K1, 9.6 \cdot 10^6, r2, 110))$	
	7.610485180×10^7	(19)
	$\rightarrow$	